

Vectors, Geometry & Similarity

Mathematics Foundations — Session 1

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Spotted a mistake?

If you find a mistake, please report it to me.

Today's Goal

- Rebuild vector intuition in $\mathbb{R}^n$, specifically in 2D/3D: addition, scalar multiplication
 - Read the dot product as a measure of alignment between two vectors
 - Use cosine similarity to compare data points
- Use in ML: rank the most similar items in a small set using cosine similarity, by hand

This class is pen-and-paper to focus on mathematics aspects. You will have Python-oriented classes.

Plan

1. Vectors in $\mathbb{R}^n$
2. First Set of Exercises
3. Dot Product and Euclidean Norm
4. Distance, Angle, Cosine Similarity
5. Applications in Machine Learning
6. Final Exercises
7. Wrap-Up

Vectors in $\mathbb{R}^n$

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Vectors: What Are They?

Vectors are a way to **store information**. For example:

- $x = (1, 0)$ is a vector of size 2
- $y = (1, 1.2, -0.5)$ is a vector of size 3
- $z = (\underbrace{1, \dots, 1}_{n \text{ times}})$ is a vector of size n

Remark: vectors can be represented in lines or in columns:

$$\underbrace{x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{size } 2 \times 1} \quad \text{or} \quad \underbrace{x = (1, 0)}_{\text{size } 1 \times 2} \quad x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Brackets or parentheses: both notations are used.

Vectors: What Are They?

If a vector is called x , then x_i is the **i -th component** of the vector.

Example: with $x = (1, 0)$ and $y = (1, 1.2, -0.5)$:

$$x_1 = 1, \quad x_2 = 0, \quad y_2 = 1.2$$

We say that a vector is of size $n \in \mathbb{N}$ if it is made of n components. We then say that $x \in \mathbb{R}^n$, where

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}}$$

What type of information can we store in a vector?

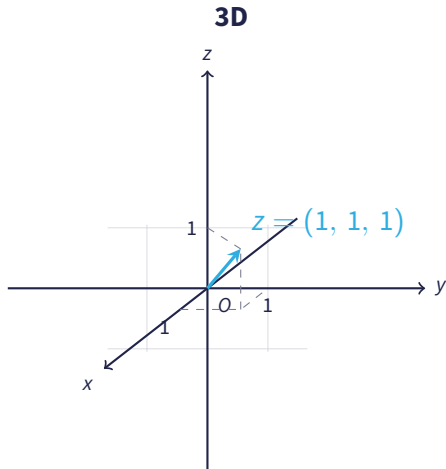
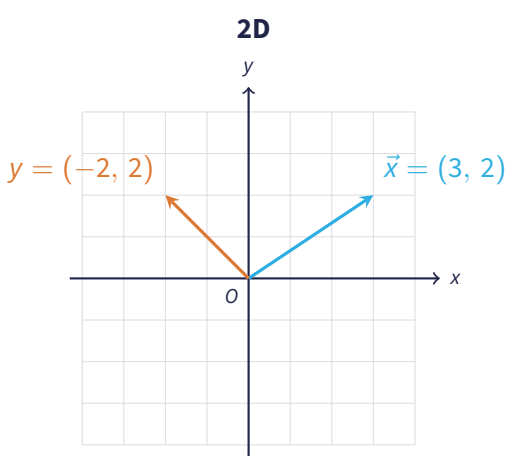
1. Coordinates
2. Words
3. Whatever you want! E.g. your grades during the semester:

$$g = (\underbrace{g_1}_{\text{first grade}}, \dots, \underbrace{g_n}_{\text{last grade}})$$

Remember

Vectors are mathematical objects that are self-defined. ML (and other areas) use them because they are useful.

Vectors: What Are They? Geometrically



Vectors size

Definition

The **size** of a vector is the number of its elements.

Example:

■ $u = (1, 1, 1, 1)$ is of size 4

■ $v = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ is of size 3

What Can We Store in a Vector?

Examples:

- 2D / 3D points
- Grades
- Stock prices
- Words
- Flower information
- Height & weight
- Model parameters in Machine Learning

Vector Operations

Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$.

Vector Addition

$$u + v = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

Scalar Multiplication, $\alpha \in \mathbb{R}$

$$\alpha u = \begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix}$$

A scalar is a 1-size element of $\mathbb{R}$: “a coefficient”.

Linear Combination

For $\lambda, \mu \in \mathbb{R}$:

$$\lambda u + \mu v = \begin{pmatrix} \lambda u_1 + \mu v_1 \\ \lambda u_2 + \mu v_2 \end{pmatrix}$$

Dot Product

$$u \cdot v = u_1 v_1 + u_2 v_2$$

Careful: it takes **two vectors** and gives a **scalar** as output.

Vector Operations: Examples

In 2D:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 0+3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \in \mathbb{R}^2$$

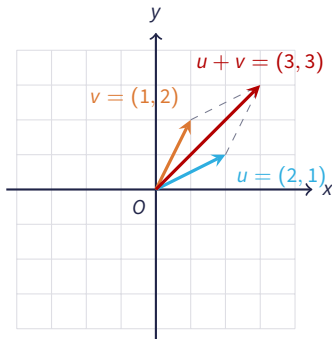
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 1 \times 2 + 0 \times 3 = \underline{2} \in \mathbb{R}$$

In 3D:

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = 1 \times 3 + 2 \times 4 + 1 \times 5 = 3 + 8 + 5 = \underline{16}$$

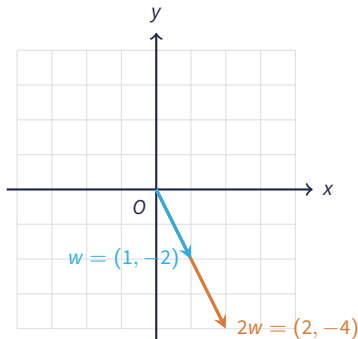
Vector Operations: Geometrical Meaning

Vector Addition



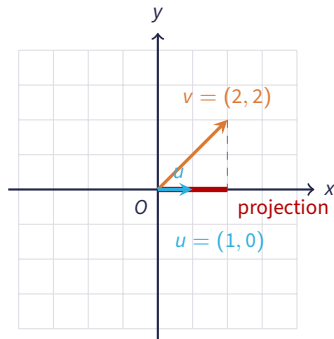
$$u + v = v + u$$

Scalar Multiplication



$2w$ has the same direction as w , twice as long

Dot Product



$$u \cdot v = 2 \times 1 + 2 \times 0 = 2$$

(projection of v on u)

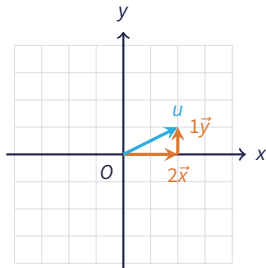
Vector Decomposition

What does $u = (2, 1)$ mean in the 2D frame $(\vec{x}, \vec{y})$?

$$u = 2 \cdot \vec{x} + 1 \cdot \vec{y}$$

Can we always do this? With 3 vectors? With 4 vectors?

3 vectors in 2D: one of them can always be built from the two others $\rightarrow$ there is one vector too many (see linear independence).



Vector Independence

Definition

Two vectors are **linearly independent** if **neither** can be written as a **nonzero multiple** of the other.

Mathematically: if we **can't** find $\lambda \in \mathbb{R}$ such that $u = \lambda v$, for $u, v \in \mathbb{R}^n$, then u and v are **linearly independent** (otherwise they are **linearly dependent**).

Example:

$$\underbrace{\begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\text{linearly dependent}}$$

but

$$\underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix} \neq \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ for all } \alpha \in \mathbb{R}}_{\text{linearly independent}}$$

Vector Independence

Corollary

Three vectors are **linearly dependent** if one can be expressed as a **linear combination** of the other two. i.e if there exist scalars α, β, γ , **not all zero**, such that

$$\alpha u + \beta v + \gamma w = 0$$

For $u, v, w \in \mathbb{R}^n$:

Either: $\exists \alpha, \beta \in \mathbb{R}$ s.t.

$$u = \alpha v + \beta w$$

Or (most general): $\exists \alpha, \beta, \gamma \in \mathbb{R}$ s.t.

$$\alpha u + \beta v + \gamma w = 0$$

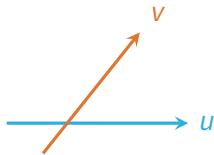
(the first form is the case where the coefficient of u is -1)

Notion of Basis

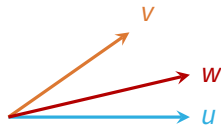
If n vectors are independent (from last definition) in a vector space of size n , then we say that these n vectors are a **basis** of this space ($\mathbb{R}^n$).

This means that these n vectors are all we need to “span” the space.

Notion of Basis



u and v are 2 vectors of size 2, clearly independent. So they are a basis of $\mathbb{R}^2$, which is the 2D space.



Since u and v are a basis of $\mathbb{R}^2$, w has to be linearly dependent with u and v , meaning there exist $\lambda, \mu \in \mathbb{R}$ such that

$$w = \lambda u + \mu v$$

Example

Example

$$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Are a and b linearly independent?

Can we find $\alpha \in \mathbb{R}$ such that $a = \alpha b$? If such an α exists:

$$a = \alpha b \iff \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3\alpha \\ \alpha \end{pmatrix} \iff \begin{cases} 1 = 3\alpha \\ 2 = \alpha \end{cases} \Rightarrow \text{impossible}$$

(Two vectors are equal if each of their components is equal.) $\Rightarrow$ **Not dependent:** a and b are linearly independent.

Problem: Linear Independence

Problem

Check if the following vectors are linearly independent:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 6 \\ 6 \\ 8 \end{bmatrix}$$

Are they linearly independent? If not, express one of them as a linear combination of the others.

We search for α, β and $\gamma \in \mathbb{R}$ such that:

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0$$

Problem: Linear Independence — Solution

$$\alpha \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 6 \\ 6 \\ 8 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} \alpha \\ 2\alpha \\ 3\alpha \end{pmatrix} + \begin{pmatrix} \beta \\ 2\beta \\ \beta \end{pmatrix} + \begin{pmatrix} 6\gamma \\ 6\gamma \\ 8\gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} \alpha + \beta + 6\gamma = 0 & L_1 \\ 2\alpha + 2\beta + 6\gamma = 0 & L_2 \\ 3\alpha + \beta + 8\gamma = 0 & L_3 \end{cases}$$

$$\Leftrightarrow \begin{cases} \alpha + \beta + 6\gamma = 0 & L_1 \leftarrow L_1 \\ -6\gamma = 0 & L_2 \leftarrow L_2 - 2L_1 \\ 2\alpha + 2\gamma = 0 & L_3 \leftarrow L_3 - L_1 \end{cases}$$

$$\Leftrightarrow \begin{cases} \alpha + \beta + 6\gamma = 0 \\ \gamma = 0 \\ 2\alpha = 0 \end{cases} \Leftrightarrow \begin{cases} \alpha = 0 \\ \beta = 0 \\ \gamma = 0 \end{cases}$$

$\Rightarrow$ The vectors are linearly independent

Problem: Linear Independence — What If?

If we had found a solution that is **not all zero**, for instance $\alpha = 2, \beta = 5, \gamma = 3$:

$$2v_1 + 5v_2 + 3v_3 = 0 \iff 2v_1 = -5v_2 - 3v_3 \iff v_1 = -\frac{5}{2}v_2 - \frac{3}{2}v_3$$

$\Rightarrow v_1$ is a linear combination of v_2 and v_3 : the vectors would be **dependent**.

First Set of Exercises

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- 2. First Set of Exercises**
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Exercises (1/2)

Exercise

Given vectors

$$a = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

1. Calculate $a + b$
2. Calculate $2a - b$
3. Are a and b linearly independent?

Exercise (1/2) – Solution

$$\mathbf{1)} \ a + b = \begin{pmatrix} 3 + 1 \\ 4 + 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$\mathbf{2)} \ 2a - b = \begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

3) Can we find $\alpha \in \mathbb{R}$ such that $a = \alpha b$?

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} \alpha \\ 2\alpha \end{pmatrix} \iff \begin{cases} 3 = \alpha \\ 4 = 2\alpha \end{cases} \Rightarrow \text{impossible } (\alpha = 3 \text{ and } \alpha = 2)$$

$\Rightarrow a$ and b are **linearly independent**.

Exercises (2/2)

Exercise

Given vectors in $\mathbb{R}^3$

$$a = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$$

1. Calculate $3a + 2b$
2. Write c as a linear combination of a and b
3. Are a , b and c linearly independent?

Exercise (2/2) – Solution

1)

$$3a + 2b = 3 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \\ -2 \end{pmatrix}$$

2) **Obvious solution.** Look at the first component: $c_1 = 2$, $a_1 = 1$ and $b_1 = 0$, so c needs $2a$:

$$2a = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}, \quad c - 2a = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = b \quad \Rightarrow \quad c = 2a + b$$

Let us now **solve the system** to find it systematically.

Exercise (2/2) — Solution

2) We search for α, β, γ such that $\alpha a + \beta b + \gamma c = 0$:

$$\begin{pmatrix} \alpha \\ 2\alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \\ -\beta \end{pmatrix} + \begin{pmatrix} 2\gamma \\ 5\gamma \\ -\gamma \end{pmatrix} = 0 \iff \begin{cases} \alpha + 2\gamma = 0 & L_1 \\ 2\alpha + \beta + 5\gamma = 0 & L_2 \\ -\beta - \gamma = 0 & L_3 \end{cases}$$

Exercise (2/2) – Solution

$$\iff \begin{cases} \alpha + 2\gamma = 0 & L_1 \\ \beta + \gamma = 0 & L_2 \leftarrow L_2 - 2L_1 \\ \beta = -\gamma & L_3 \end{cases}$$

$$\iff \begin{cases} \alpha = -2\gamma \\ \beta = -\gamma \end{cases}$$

$\Rightarrow$ This can work for different values: we can choose a value.

For instance, $\beta = 1$ gives (we recover the obvious solution):

$$\gamma = -\beta = -1, \quad \alpha = 2$$

$$\iff 2a + 1b - c = 0 \iff c = 2a + b$$

Check:

$$\begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = c$$

3) $\Rightarrow a, b$ and c are **linearly dependent**.

Dot Product and Euclidean Norm

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Dot Product Operation

Recall: the dot product of 2 vectors is the following:

$$u \cdot v = u_1 \cdot v_1 + u_2 \cdot v_2$$

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad \longrightarrow \quad u_1 v_1 + u_2 v_2$$

Multiply the components two by two, then sum.

Dot Product Operation

$$a = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \quad a \cdot b = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32$$

Definition

If, for a and b , we have $a \cdot b = 0$, we say that a and b are orthogonal, or that a is orthogonal to b . We can write $a \perp b$.

Dot Product and Independence in 2D

$$a = (0, 2)$$



$$b = (1, 0)$$



$$a \cdot b = 0 \times 1 + 2 \times 0 = 0$$

$$b = (-1, 1)$$



$$a = (1, 1)$$



$$a \cdot b = 1 \times (-1) + 1 \times 1 = 0$$

$$a = (2, 2)$$



$$b = (2, 1)$$



$$a \cdot b = 2 \times 2 + 2 \times 1 = 6$$

Orthogonal vectors ($a \cdot b = 0$) in the first two cases; not in the last one.

Dot Product Properties

$$a \cdot b = b \cdot a \quad (\text{Symmetry})$$

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad (\text{Bilinearity})$$

Euclidean Norm : intuition

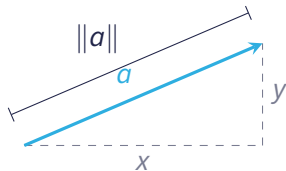
What happens if we do $a \cdot a$, with $a = \begin{pmatrix} x \\ y \end{pmatrix}$?

$$a \cdot a = x \times x + y \times y = x^2 + y^2 \quad ("a^2")$$

This is written:

$$\|a\|^2 = x^2 + y^2$$

$\|a\|$ is the **norm** of a : its “length”.



Euclidean Norm : Definition

We also write:

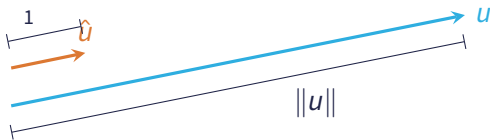
$$\|a\| = \sqrt{a \cdot a}$$

$\|a\|$: **the norm of** a

Normalization and Unit Vectors

In Machine Learning (and in maths), we like vectors of **length 1**, and we can create any “unit-size” vector by doing:

$$\hat{u} = \frac{u}{\|u\|}$$



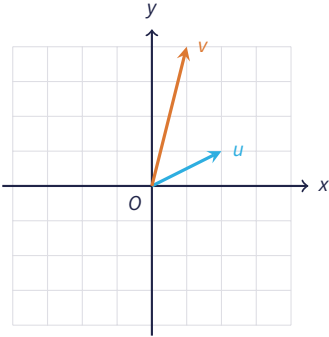
Basis (Again)

Recall: if 2 vectors (u, v) are linearly independent, they are a basis of $\mathbb{R}^2$, and:

- if $u \cdot v = 0$, (u, v) is said to be an **orthogonal** basis;
- if $u \cdot v = 0$ and $\|u\| = 1$, $\|v\| = 1$, (u, v) is said to be an **orthonormal** basis.

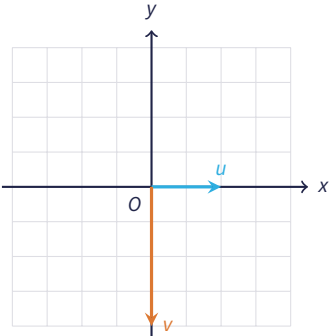
Orthogonal and Orthonormal Basis

Neither



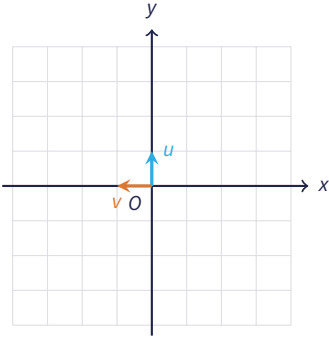
$$u \cdot v \neq 0$$

Orthogonal



$$u \cdot v = 0$$

Orthonormal



$$u \cdot v = 0, \|u\| = \|v\| = 1$$

Norms (General Case)

Norm is a larger concept:

Definition: norm

A norm on $\mathbb{R}^n$ is a function $\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}$ such that, for all $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

1. $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$
2. $\|\lambda x\| = |\lambda| \|x\|$ where $|\lambda| = \lambda$ if $\lambda \geq 0$, $|\lambda| = -\lambda$ if $\lambda < 0$
3. $\|x + y\| \leq \|x\| + \|y\|$

The norms which can be expressed with a scalar product,

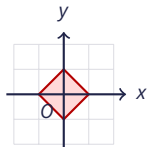
$$\|u\| = \sqrt{u \cdot u},$$

are said to be **Euclidean**.

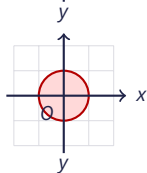
Norms (General Case)

For $x = (x_1, x_2)$, the following norms are used in Machine Learning. **Unit circle: points of length 1**

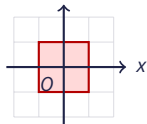
L^1 $\|x\|_1 = |x_1| + |x_2|$ (absolute value, see before)



L^2 $\|x\|_2 = \sqrt{x_1^2 + x_2^2} = \sqrt{x_1x_1 + x_2x_2}$



L^∞ $\|x\|_\infty = \max_{i=1\dots n} |x_i|$



Normalization and Unit Vectors

Normalization $\longrightarrow$ the fact of dividing a vector by its norm:

$$\hat{u} = \frac{u}{\|u\|}$$

Exercises: Norms (1/2)

Exercise

Let $x = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$. Calculate $\|x\|_1$, $\|x\|_2$ and $\|x\|_\infty$.

$$\|x\|_1 = |x_1| + |x_2| = 3 + 4 = 7$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\|x\|_\infty = \max(|x_1|, |x_2|) = |-4| = 4$$

Exercises: Norms (2/2)

Exercise

Let $v = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$. Calculate $\|v\|_2$, then normalize v into a unit vector $\hat{v}$ and check that $\|\hat{v}\|_2 = 1$.

Exercises: Norms (2/2) — Solution

$$\|v\|_2 = \sqrt{2^2 + (-2)^2 + 1^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

$$\hat{v} = \frac{v}{\|v\|_2} = \frac{1}{3} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2/3 \\ -2/3 \\ 1/3 \end{pmatrix}$$

$$\|\hat{v}\|_2 = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = \sqrt{\frac{9}{9}} = 1 \quad \checkmark$$

Distance, Angle, Cosine Similarity

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Link with ML

All information in ML is stored in **vectors**.

And we want to **compare** this information:

⇒ **Distance, Angle, Cosine Similarity**

Distance

The **distance** between two vectors a and b is defined by

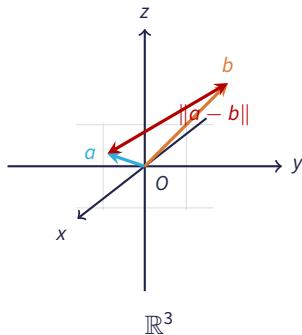
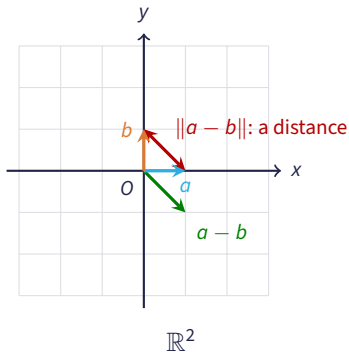
$$d(a, b) = \underbrace{\|a - b\|}_{\text{the norm of } a - b}$$

Note:

$$d(a, b) = d(b, a) = \|b - a\|$$

Distance in Geometry

$$a = (1, 0), \quad b = (0, 1)$$



$$d(a, b) = \|a - b\| = \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\| = \left\| \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Distance in Geometry

$$\text{In } \mathbb{R}^3: \quad a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$d(a, b) = \|a - b\|_2 = \left\| \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{pmatrix} \right\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

Careful

The **2** comes from the **norm** (Euclidean norm, L^2), not from the dimension: here the space is $\mathbb{R}^3$, and the same 2 appears in $\mathbb{R}^2$.

Angle (1/5)

We've seen that:

$$u \cdot v = 0 \iff u \perp v$$

We also know that if u and v are aligned, then $\exists \lambda$ such that $u = \lambda v$, which means that (for $\lambda > 0$):

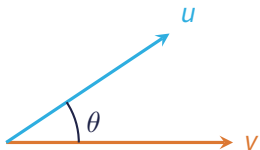
$$u \cdot v = (\lambda v) \cdot v = \lambda \times v \cdot v = \lambda \|v\|^2 = \lambda \|v\| \cdot \|v\| = \|u\| \|v\|$$

Angle (2/5)

⇒ These are part of a larger concept, which is the **angle** between u and v :

We have:

$$u \cdot v = \|u\| \|v\| \cos(u, v)$$

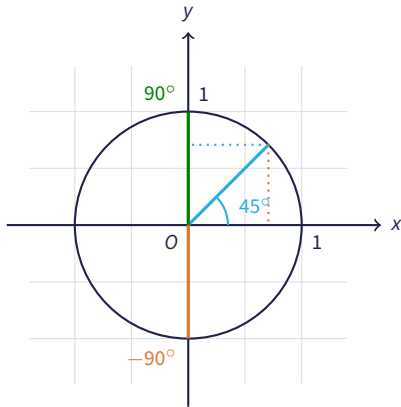


Key formula

$$u \cdot v = \|u\| \|v\| \cos(\theta)$$

Angle (3/5)

What is the **cosine** / **sine** of an angle? (Unit circle)



$$\sin(45^\circ) = \frac{\sqrt{2}}{2} \approx 0.7$$

$$\cos(45^\circ) = \frac{\sqrt{2}}{2} \approx \frac{1.41}{2} \approx 0.7$$

$$\cos(90^\circ) = 0, \quad \sin(90^\circ) = 1$$

$$-90^\circ \text{ (or } 270^\circ\text{): } \cos(-90^\circ) = 0, \\ \sin(-90^\circ) = -1$$

Angle (4/5)

Example: we have $u \cdot v = \|u\| \|v\| \cos \theta$, with

$\cos \theta = +1$ if vectors are aligned



$\cos \theta = 0$ if vectors are $\perp$



$\cos \theta = -1$ if vectors are in opposite directions



Angle (5/5)

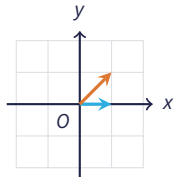
The angle is defined by:

$$\theta = \arccos \left(\frac{u \cdot v}{\|u\| \|v\|} \right)$$

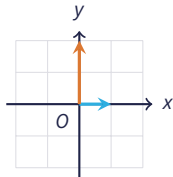
Remark: it is way easier to compute if $\|u\| = 1$ and $\|v\| = 1$:

$$\theta = \arccos(u \cdot v)$$

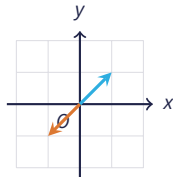
Angle Examples in 2D and 3D



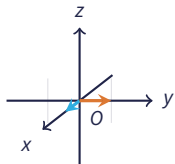
2D: $(1, 0)$ and $(1, 1)$
 $\cos \theta = \frac{1}{\sqrt{2}}, \theta = 45^\circ$



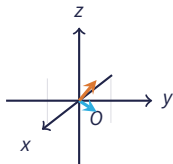
2D: $(1, 0)$ and $(0, 2)$
 $\cos \theta = 0, \theta = 90^\circ$



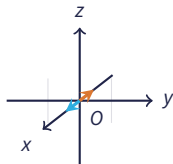
2D: $(1, 1)$ and $(-1, -1)$
 $\cos \theta = -1, \theta = 180^\circ$



3D: $(1, 0, 0)$ and $(0, 1, 0)$



3D: $(1, 1, 0)$ and $(1, 1, 1)$



3D: $(1, 0, 0)$ and $(-1, 0, 0)$

Cosine Similarity

Cosine similarity is just the cosine of the angle between 2 vectors:

Cosine similarity

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

⇒ Tells you **how similar** u and v are to one another.

Link Between the Dot Product, Norm and Distance

- Norm from the dot product: $\|a\| = \sqrt{a \cdot a}$
- Distance from the norm: $d(a, b) = \|a - b\|$

Applications in Machine Learning

1. Vectors in $\mathbb{R}^n$
2. First Set of Exercises
3. Dot Product and Euclidean Norm
4. Distance, Angle, Cosine Similarity
- 5. Applications in Machine Learning**
6. Final Exercises
7. Wrap-Up

Link with ML: Cosine Similarity

Iris flowers: 4 measurements per flower (cm, rounded). Which species is the mystery flower q closest to? Each flower is a vector of $\mathbb{R}^4$, e.g. $s = (5, 3, 1, 0)$.

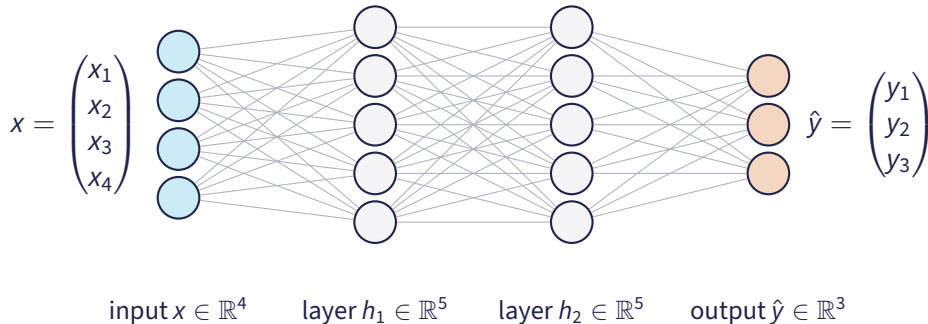
	sepal length	sepal width	petal length	petal width
setosa s	5	3	1	0
versicolor c	6	3	4	1
virginica g	7	3	6	2
mystery q	6	3	5	2

Cosine with q : closest to virginica, setosa is the farthest

$$\cos(q, s) = \frac{44}{\sqrt{74}\sqrt{35}} \approx 0.86 \quad \cos(q, c) = \frac{67}{\sqrt{74}\sqrt{62}} \approx 0.99 \quad \cos(q, g) = \frac{85}{\sqrt{74}\sqrt{98}} \approx 1.00$$

Link with ML: Deep Learning

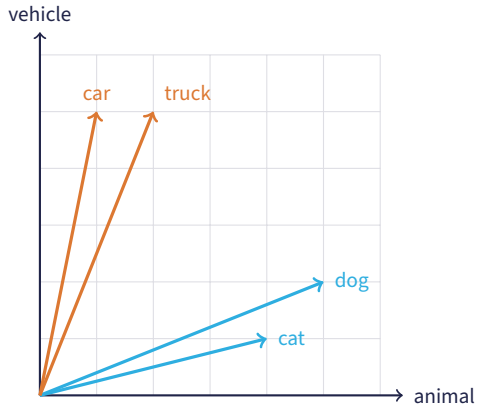
Neural network:



Key idea

Vectors everywhere! Inputs, hidden layers and outputs of a neural network are all vectors.

Link with LLMs: Word Embeddings



Each word $\rightarrow$ a vector in $\mathbb{R}^2$ (axes: “animal-ness”, “vehicle-ness”).

word	animal	vehicle
cat	4	1
dog	5	2
car	1	5
truck	2	5

Real embeddings (word2vec, GloVe, LLMs):
hundreds to thousands of dimensions,
learned from text, not hand-designed.

Link with LLMs: Similarity Between Words

$$\cos(\text{cat}, \text{dog}) = \frac{4 \cdot 5 + 1 \cdot 2}{\sqrt{17} \sqrt{29}} \approx 0.99$$

$$\cos(\text{cat}, \text{car}) = \frac{4 \cdot 1 + 1 \cdot 5}{\sqrt{17} \sqrt{26}} \approx 0.43$$

	cat	dog	car	truck
cat	1	0.99	0.43	0.59
dog	0.99	1	0.55	0.69
car	0.43	0.55	1	0.98
truck	0.59	0.69	0.98	1

Reading

cat/dog and car/truck are nearly aligned (cosine close to 1): similar words point in similar directions.

Final Exercises

1. Vectors in $\mathbb{R}^n$
2. First Set of Exercises
3. Dot Product and Euclidean Norm
4. Distance, Angle, Cosine Similarity
5. Applications in Machine Learning
- 6. Final Exercises**
7. Wrap-Up

Exercises (1/3)

Exercise

Given vectors

$$x = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

1. Calculate the Euclidean norm $\|x\|$
2. Calculate the Euclidean distance between x and y
3. Calculate the cosine similarity between x and y

Exercises (1/3) — Solution

1) $\|x\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

2) $x - y = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, so $d(x, y) = \|x - y\| = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5} \approx 4.47$

3) $x \cdot y = 4 \times 0 + 3 \times 5 = 15$, $\|y\| = 5$:

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \|y\|} = \frac{15}{5 \times 5} = \frac{3}{5} = 0.6$$

Exercises (2/3)

Exercise

Given vectors

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1. Calculate the dot product $u \cdot v$
2. Calculate the norms $\|u\|$ and $\|v\|$
3. Calculate the cosine similarity between u and v

Exercises (2/3) — Solution

1) $u \cdot v = 1 \times 1 + 0 \times 1 = 1$

2) $\|u\| = \sqrt{1^2 + 0^2} = 1$, $\|v\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

3)

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \|v\|} = \frac{1}{1 \times \sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.71 \quad (\theta = 45^\circ)$$

Exercises (3/3)

Exercise

Given a reference vector r and two candidates p_1, p_2 :

$$r = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad p_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad p_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

1. Calculate the cosine similarity between r and p_1
2. Calculate the cosine similarity between r and p_2
3. Which candidate is more similar to r ?

Exercises (3/3) — Solution

First, $\|r\| = \sqrt{5}$ and $\|p_1\| = \|p_2\| = \sqrt{10}$.

1) $r \cdot p_1 = 2 \times 1 + 1 \times 3 = 5:$

$$\cos(r, p_1) = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \approx 0.71$$

2) $r \cdot p_2 = 2 \times 3 + 1 \times 1 = 7:$

$$\cos(r, p_2) = \frac{7}{\sqrt{50}} \approx 0.99$$

3) p_2 is more similar to r (its cosine is larger, so the angle is smaller).

Bonus Exercises (1/3)

Bonus 1: normalize the Iris flowers

Using the Iris vectors $s = (5, 3, 1, 0)$, $c = (6, 3, 4, 1)$, $g = (7, 3, 6, 2)$, $q = (6, 3, 5, 2)$:

1. Normalize each vector into a unit vector $\hat{s}$, $\hat{c}$, $\hat{g}$, $\hat{q}$.
2. Calculate $\hat{q} \cdot \hat{s}$, $\hat{q} \cdot \hat{c}$ and $\hat{q} \cdot \hat{g}$.
3. Compare with the cosine similarities computed on the Iris slide (“Link with ML”). What do you notice?

Bonus 1 — Solution

vector	norm	unit vector
$s = (5, 3, 1, 0)$	$\sqrt{35} \approx 5.92$	$\hat{s} \approx (0.85, 0.51, 0.17, 0)$
$c = (6, 3, 4, 1)$	$\sqrt{62} \approx 7.87$	$\hat{c} \approx (0.76, 0.38, 0.51, 0.13)$
$g = (7, 3, 6, 2)$	$\sqrt{98} \approx 9.90$	$\hat{g} \approx (0.71, 0.30, 0.61, 0.20)$
$q = (6, 3, 5, 2)$	$\sqrt{74} \approx 8.60$	$\hat{q} \approx (0.70, 0.35, 0.58, 0.23)$

$$\hat{q} \cdot \hat{s} \approx 0.86, \quad \hat{q} \cdot \hat{c} \approx 0.99, \quad \hat{q} \cdot \hat{g} \approx 1.00$$

These are exactly the cosine similarities of the Iris slide: for **unit** vectors, the dot product *is* the cosine similarity.

Bonus Exercises (2/3)

Bonus 2: word similarity table

Toy word vectors (animal, vehicle): cat = (4, 1), dog = (5, 2), car = (1, 5), truck = (2, 5).
Fill in the cosine similarity table.

	cat	dog	car	truck
cat	1			
dog		1		
car			1	
truck				1

Bonus 2 — Solution

	cat	dog	car	truck
cat	1	0.99	0.43	0.59
dog	0.99	1	0.55	0.69
car	0.43	0.55	1	0.98
truck	0.59	0.69	0.98	1

The table is symmetric ($\cos(u, v) = \cos(v, u)$) with 1 on the diagonal. For example

$$\cos(\text{cat}, \text{dog}) = \frac{4 \cdot 5 + 1 \cdot 2}{\sqrt{17}\sqrt{29}} \approx 0.99.$$

Bonus Exercises (3/3)

Bonus 3: a new word

A new word $\text{bus} = (1, 4)$ is added to the same space.

1. Calculate its cosine similarity with cat, dog, car and truck.
2. Which word is it closest to?
3. Is that what you expected? Why?

Bonus 3 — Solution

1) With $\text{bus} = (1, 4)$ and $\|\text{bus}\| = \sqrt{17}$:

$$\cos(\text{bus}, \text{cat}) = \frac{1 \cdot 4 + 4 \cdot 1}{\sqrt{17}\sqrt{17}} = \frac{8}{17} \approx 0.47$$

$$\cos(\text{bus}, \text{dog}) = \frac{1 \cdot 5 + 4 \cdot 2}{\sqrt{17}\sqrt{29}} \approx 0.59$$

$$\cos(\text{bus}, \text{car}) = \frac{1 \cdot 1 + 4 \cdot 5}{\sqrt{17}\sqrt{26}} \approx 1.00$$

$$\cos(\text{bus}, \text{truck}) = \frac{1 \cdot 2 + 4 \cdot 5}{\sqrt{17}\sqrt{29}} \approx 0.99$$

2) Closest word: **car** (truck is a very close second).

3) Expected: a bus is a vehicle, so its vector has a high “vehicle-ness” and a low “animal-ness”, like car and truck and unlike cat and dog.

Wrap-Up

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Recap

1. Vectors **store information**.
2. In $\mathbb{R}^n$, they are represented in a **basis** of size n (n independent vectors).
3. The **norm** is the length of a vector.
4. **Cosine similarity** tells you how close two vectors are.

Next session teaser

Matrices are stacked vectors:

$$M = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \quad (\text{columns: } \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 2 \\ 2 \end{pmatrix})$$

Re-do the exercises, ask me question if needed.

Class Feedback

- Was this class too fast? Too slow?
- Easy to follow? Hard to follow?
- Boring?

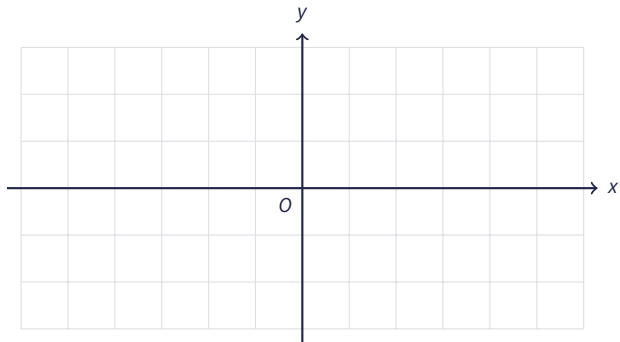
Anonymous feedback here: [feedback form](#)

Or email me: etienne.chassaing.conseil@gmail.com

Appendix

8. Appendix

Coordinate Grid



Backup: Word Analogies (Toy Example)

Toy vectors in $\mathbb{R}^4$, axes: royalty, male, female, animal.

word	royalty	male	female	animal
king	9	9	1	0
queen	9	1	9	0
man	1	9	1	0
woman	1	1	9	0
cat	0	4	4	9

$$\text{king} - \text{man} + \text{woman} = (9, 1, 9, 0) = \text{queen}$$

Cosine with the result

queen 1.00 woman 0.78 king 0.61 cat 0.30 man 0.23